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Phosphorylation promotes Al(III) binding to proteins: GEGEGSGG as a case study†

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Aluminum, the third most abundant element in the Earth's crust and one of the key industrial components of our everyday life, has been associated with several neurodegenerative diseases due to its ability to promote neurofilament tangles and β -amyloid peptide aggregation. However, the experimental characterization of aluminum speciation *in vivo* is a difficult task. In the present study, we develop a theoretical protocol that combines molecular dynamics simulations, clustering of structures, and density functional theory for the characterization of the binding of aluminum to the synthetic neurofilament analogue octapeptide GEGEGSGG and its phosphorylated variant. Our protocol is tested with respect to previous NMR experimental data, which allows for a full interpretation of the experimental information available and its relationship with key thermodynamic quantities. Our results demonstrate the importance of phosphorylation in the ability of a peptide to bind to aluminum. Thus, phosphorylation: (i) changes the binding pattern of aluminum to GEGEGSGG, shifting the preferential binding site from the C-terminal to S6(P); (ii) increases the binding affinity by a factor of around 15 kcal mol⁻¹ in free energy; and (iii) may cause significant changes in the secondary structure and stiffness of the polypeptide chain, specially in the case of bidentate binding modes. Our results shed light on the possibility of aluminum to induce aggregation of β -amyloid proteins and neurofilament tangles.

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1. Introduction

Although aluminum is the third most abundant element in the Earth's crust, it is not an essential element in any living system and it has been excluded from biotic cycles.^{1,2} However, over the last century and because of its various technological applications, human intervention has increased the bioavailability of the Al(III) cation. Unfortunately, this is unlikely to be without consequences. There is increasing evidence of the potential toxic effects of aluminum in biological systems,^{3–5} linking the presence of aluminum in the human body with several diseases,^{6,7} such as dialysis encephalopathy, osteodystrophy and microcytic anemia derived from chronic renal failure wherein the aluminum accumulates in the kidney, liver, bone and heart.^{8,9} The full molecular basis of aluminum toxicity are still unknown, but some potential mechanisms have been outlined. For instance, it has been well established that aluminum shows a significant pro-oxidant activity in biological systems,^{10–12} it also inhibits the normal function of various enzymes involved in the glycolysis pathway^{13–16} and in the production of glutamate^{17,18} affecting

the TCA cycle.^{19–21} Moreover, several studies have shown that aluminum competes with magnesium as a metal-ATP co-factor^{22–27} and that it can modify the electronic and structural properties of the chelated biological ligands.^{28–30} In general, aluminum has a predominance for O donor species,³¹ with negatively charged groups, such as carboxylates, phenolates, catecholates and phosphates, because they are the strongest aluminum binding biomolecules; phosphates are particularly important because of the diverse range of molecules that contain this group and the cellular processes in which they participate.^{32–34}

Aluminum is considered as a neurotoxic element associated with Alzheimer's disease (AD), but it is not clear the role that aluminium plays in neurodegeneration.^{35–39} In a seminal study, Exley *et al.* demonstrated the ability of aluminum to bind β -amyloids,⁴⁰ which leads to a switch in structure between the α -helix and β -sheet. Moreover, *in vitro* experiments determined that Al(III) promotes the aggregation of β -amyloid peptides more efficiently than other metals,^{41–43} with the main constituent of insoluble amyloid plaques known as senile plaques, and in fact, aluminum has been detected in senile plaques extracted from the brains of patients with AD.⁴⁴ In addition, this metal also induces the abnormal neurofilament tangle (NFT) aggregation and promotes the hyperphosphorylation of normal proteins.^{45–47} As a matter of fact, phosphorylation of peptides is thought to increase their affinity for aluminum and could be a key aspect

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in the understanding the aluminum promotion of the NFT aggregation process. Therefore, there is great interest to understand the changes that phosphorylation causes in the interaction of polypeptides with aluminum. As a paradigmatic case, Hollender *et al.* performed a full study of the structure of the synthetic neurofilament analogue octapeptide GEGEGSGG and its Ser(P) derivative, both in the absence and presence of aluminum.⁴⁸ They used potentiometry, CD and multinuclear NMR spectroscopy, and the results were complemented with MD simulations, albeit of limited accuracy, due to the lack of proper aluminum parameters available at the time and the inherent limitations of a classical force field. Quite interestingly, they observed that both non-phosphorylated and phosphorylated octapeptide were able to bind to Al(III). However, the NMR chemical shifts and broadening of the specific signals suggested a significant change in the Al-binding pattern upon phosphorylation. Thus, in the non-phosphorylated polypeptide, coordination of aluminum to the

C-terminal carboxylate is preferred, with the E4-sidechain as an additional possible chelation site. However, no evidence was found for coordination to the Ser alcoholic-OH group, whereas in the case of the phosphorylated derivative, the Ser6(P) behaved as the primary binding site, with the C-terminal carboxylate as a potential additional binder. Their results point clearly to an increase in the aluminum affinity of a given residue upon phosphorylation, which could provoke significant conformational changes and a more pronounced ability to promote protein aggregation processes.

Given the complexity of the coordination chemistry of aluminum in biological systems, theoretical calculations can help elucidate the structural and electronic aspects of the aluminum complexes in solution.^{49–56} In the present study, we apply a protocol that combines molecular dynamic simulations with properly optimized Al(III) parameters,⁵⁷ density functional theory (DFT) calculations of the NMR shielding constants, and DFT

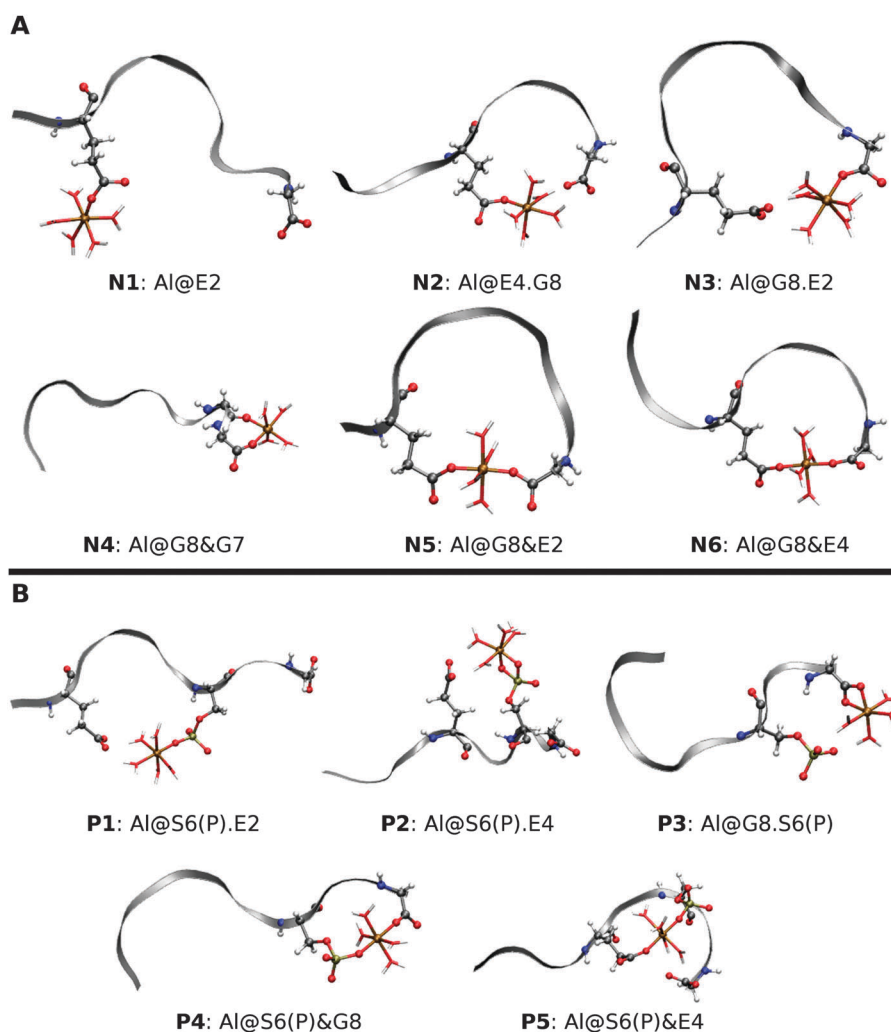


Fig. 1 The structures considered for non-phosphorylated (**Ni**, $i = 1-6$) and phosphorylated (**Pi**, $i = 1-5$) peptides. In the non-phosphorylated peptide (**A**), the ion is bound to the glutamic acids E2 or E4 (**N1** and **N2**), to the C-terminal via a mono and bidentate fashion (**N3** and **N4**) and to the C-terminal in combination with E2 or E4 (**N5** and **N6**). In the phosphorylated derivative (**B**), the ion is coordinated to the phosphate group via a mono and bidentate manner (**P1** and **P2**), to the C-terminal (**P3**) and to the phosphate group combined either with the C-terminal (**P4**) or with E4 (**P5**). The labels indicate the residue or residues at which the Al(III) is coordinated and the "dot" stands for the group placed in the second coordination sphere.

estimations of the binding affinities in the context of polarizable continuum models to consider the bulk solvent effects, for both the phosphorylated and non-phosphorylated GEGEGSGG octapeptide, and in the presence/absence of Al(III). Different coordination modes were tested (see Fig. 1). The aim of this study was to understand the changes in the polypeptide structure and electronic properties caused by the presence of Al(III) in combination with peptide phosphorylation. A careful comparison of our results with the experimental NMR chemical shifts showed a good agreement and allowed one to rationalize the interaction mode of aluminum towards phosphorylated amino acids. Finally, we report the calculation of key thermodynamic data for complex formation, which allowed us to estimate the increase in aluminum binding affinity expected by phosphorylation of Ser in the octapeptide and we compare our estimates with previous studies using similar methodologies of well-known aluminum binders in biological media such as citrate, 2,3-DPG or ATP. Our results point to a clear increase in binding affinity upon peptide phosphorylation, although its degree is not high enough as to be able to compete with strong aluminum chelators.

2. Methodology

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The theoretical protocol used throughout this study is summarized in Scheme 1. Each of the structures in Fig. 1 were **built from scratch with Molden**⁵⁸ in a β -conformation with the N-terminal protonated and the C-terminal deprotonated as well as the glutamic acids. The MD trajectories were carried out using Gromacs package,^{59,60} version 4.5.7 and the CHARMM36 force field,⁶¹ including the non-bonded parameters of Al(III) from Faro *et al.*, used previously.^{57,62,63} The peptides were solvated in a 3.75 nm³ dodecahedral water box with the TIP3P model,⁶⁴ depending on the case, sodium and chloride ions were added

to neutralize the charge. An initial energy minimization of the systems was performed using the steepest descent method for 10 000 steps. The Particle Mesh Ewald (PME) method^{65,66} was used to calculate the electrostatic interactions with a cut-off of 0.8 nm for the short range van der Waals interactions. The integration time step was of 2 fs and after 200 ps equilibration, trajectories of 20 ns were performed in the canonical ensemble at 300 K with the Bussi thermostat.⁶⁷ All MD trajectories were sampled every 10 ps and the structures were clustered by the **Gromacs** utility that uses the algorithm described by Daura *et al.*⁶⁸ In our case, this clustering brings together more than 46% of the sampled structures into the first five groups (Table S1, ESI[†]).

The NMR calculations were carried out using one representative structure of each group, including only explicit water molecules located at ~ 3.0 Å from the peptide; the values were averaged to maintain the dynamic effects and compared with the experimental data. The NMR chemical shifts with respect to sodium 2,2-dimethyl-2-silapentane-5-sulfonate (DSS) were computed with DFT as implemented in the deMon2k code,^{69–72} using the exchange-correlation functional PBE,⁷³ the orbital basis set aug-cc-pVTZ^{74,75} and an automatically generated GEN-A2 auxiliary basis set, used to avoid the evaluation of four center integrals by invoking the variational fitting of the Coulomb energy.^{76,77}

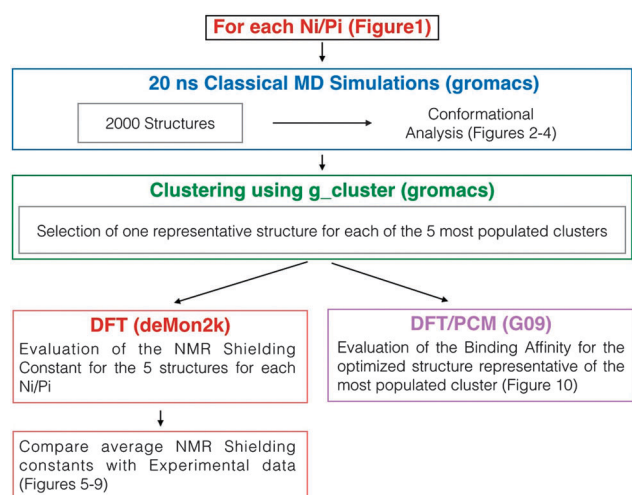
The binding affinities were computed with the Gaussian09 program;⁷⁸ one structure from the most populated cluster of each coordination mode was optimized and characterized as local minima with a harmonic frequency analysis using the B3LYP^{79,80} functional and the 6-31+G(d) basis set, the zero-point vibrational energy (ZPVE) and the thermal vibrational corrections at 298 K to evaluate the enthalpies and Gibbs free energies were extracted from the frequency analysis; the entropy was computed using the standard statistical mechanics expressions for the partition function in the canonical ensemble. Solvent effects were included using the implicit solvation model IEFPCM.⁸¹ Electronic energies were refined using single-point energy calculation at the B3LYP/6-311++G(d,p) level.

3. Results

3.1. Conformational analysis from MD simulations

The Ramachandran plots obtained from the MD simulations for all the structures can be found in Fig. 2. In the absence of aluminum, there are well-defined regions that belongs to a β -sheet conformation (-135° , 135°), to the PPII (-75° , 150°) and α -helix regions both left- (-70° , 25°) and right-handed (-70° , -30°), although the starting structures were created using a β -sheet conformation. This suggests that our 20 ns MD simulation time allows for an appropriate sampling of the conformational space. The phosphorylation of S6 has a sizeable effect in the α -helix regions, being less populated upon phosphorylation.

The presence of aluminum can also have a significant effect on the conformational preference depending on the coordination type. In the case of the non-phosphorylated peptides, the largest



Scheme 1 The theoretical protocol used throughout this study for each of the structures shown in Fig. 1 based on: (i) MD simulations, (ii) clustering of structures (iii) DFT evaluation of the NMR shielding constants for selected structures and (iv) DFT/PCM evaluation of the binding free energy/enthalpy affinities.

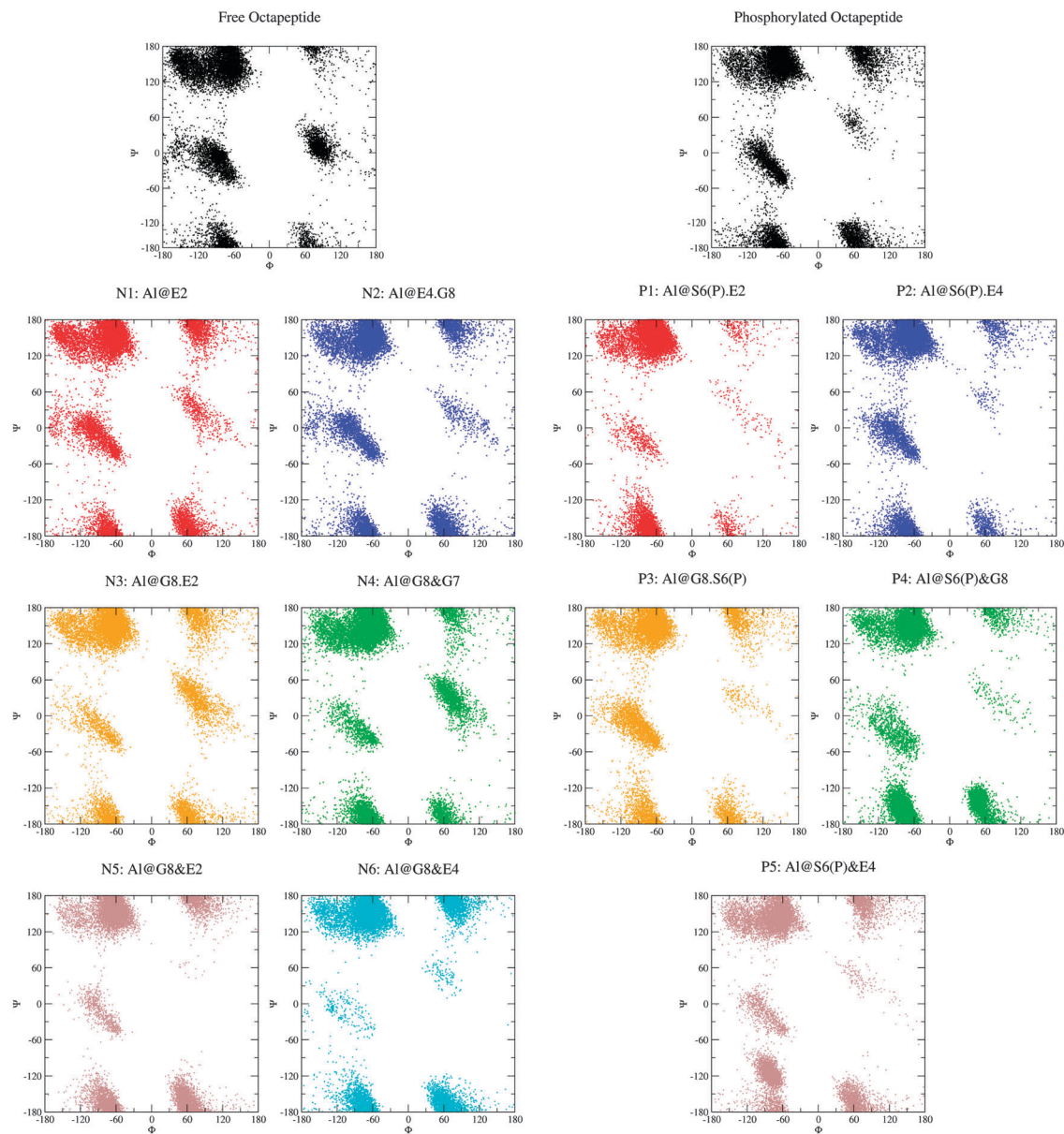


Fig. 2 The Ramachandran plots obtained from the MD simulations of the non-phosphorylated and phosphorylated structures.

effects are observed for the two bidentate structures (N5 and N6), wherein the α region is very poorly populated, and correspondingly, there is a stabilization of the β -sheet conformation. The rest of the possible aluminum coordination modes have a lower effect in the β/α conformational preference. In the case of the phosphorylated structures, we observe a lower tendency to occupy the α region as well, although in this case, the free phosphorylated octapeptide also shows a depletion of this zone.

The presence of aluminum can also lead to changes in the rigidity of the polypeptide. In Fig. 3, the root mean square fluctuations (RMSF) of $C\alpha$'s as a function of the residue are depicted. In most of them we observe a lowering of the RMSF's with respect to the free peptide, especially for N5, N6, P4 and P5, all of them, bidentate structures. This implies

that the aluminum has the ability to increase the rigidity of the octapeptide by decreasing the mobility of the coordination site.

To further assess the changes in the structure upon aluminum binding, we depict the average distance between a pair of residues, comparing the results for aluminum complexes (y axis) *versus* the free octapeptides (x axis), as shown in Fig. 4. Data above the diagonal line implies an elongation with respect to the distance in the free octapeptide. Conversely, the points below the diagonal line imply a shortening of the distance between two residues upon aluminum binding. A variety of situations are found depending on the type of aluminum coordination, demonstrating that the secondary structure of the peptide is sensitive to aluminum binding. For instance, in the case of the non-phosphorylated octapeptide the bidentate

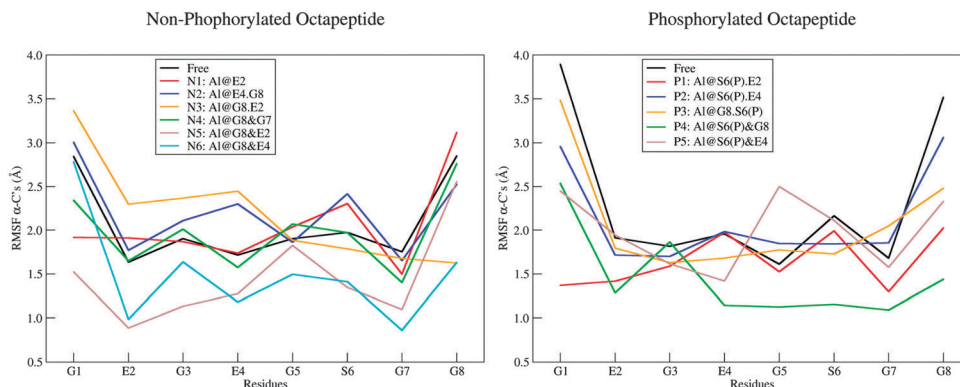


Fig. 3 The root mean square fluctuations of $C\alpha$'s in each of the non-phosphorylated and phosphorylated structures, as calculated from the MD simulations.

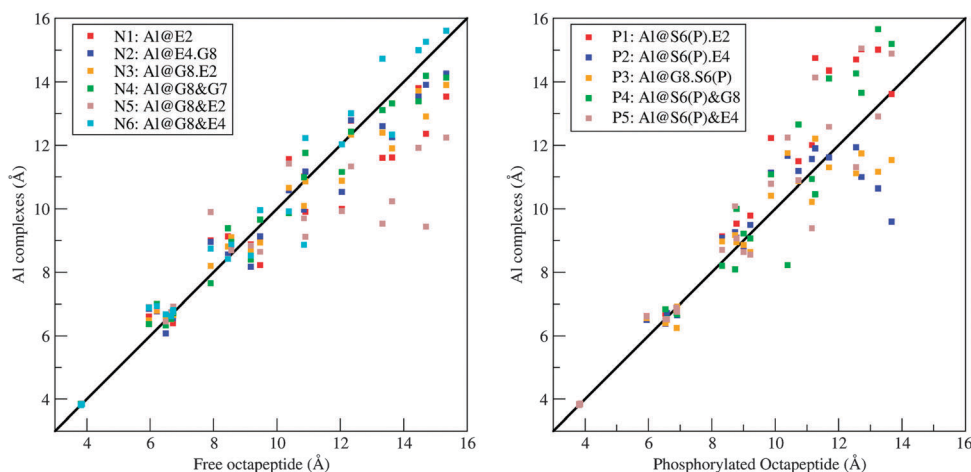


Fig. 4 The coordinating effect of aluminum on the average distances between pairs of $C\alpha$'s in each of the non-phosphorylated and phosphorylated structures, as calculated from the MD simulations.

binding to G8 and E2 (N5) leads to a clear shortening of the distances between residues. The same is true for the phosphorylated P2 structure. However, the binding to S6(P) in P1 leads to an elongation of the structure of the polypeptide.

3.2. NMR chemical shifts

In this section, the NMR chemical shifts are discussed. It is worth noting that the computed data are the average of five structures from the MD simulations of each coordination mode and of the free octapeptides both non-phosphorylated and phosphorylated. First, we compare the data obtained for the free octapeptides with the experimental values to gain an idea of the differences with the methodology used (Section 3.2.1) and subsequently, the corresponding results for the non-phosphorylated complexes are discussed (Section 3.2.2) to evaluate the effect of the aluminum coordination in the peptide; the signals of the complexes are compared with the free octapeptide to visualize the chemical shifts observed experimentally. Finally, the same was carried out for the phosphorylated complexes. Moreover, the ^{31}P signals are also compared (Section 3.2.3).

3.2.1. NMR chemical shifts of the GEGEGSGG peptide and its Ser(P) derivative. The computed chemical shifts of the free octapeptide are in good agreement with the experimental data. With respect to the ^1H NMR experimental values, the largest deviations in the CH and CH_2 groups are 0.6 ppm, whereas the amide protons have differences less than 3.0 ppm (Fig. 5A and B). The ^{13}C NMR chemical shifts for the CH and CH_2 groups have maximum differences of 5.0 and 12.1 ppm, respectively, and the CO and COO groups deviate from the experimental values up to 8.6 and 6.5 ppm, respectively (Fig. 5C and D).

The ^1H NMR chemical shifts of the phosphorylated peptide are quite similar to the non-phosphorylated peptide, which was also observed experimentally;⁴⁸ the variation between the computed values are less than 1.0 ppm, whereas in the ^{13}C NMR chemical shift values, the general trend is in good agreement with the experimental data with larger differences of 7.6 ppm for the CH signals, 15.0 ppm for the CH_2 signals, 8.3 ppm for the CO signals and 7.1 ppm for the COO signals (Fig. 6). It is worth noting that the ^{13}C NMR differences in the chemical shifts obtained with the selected methodology were in most cases twice that expected when the optimized structures were used.^{72,82–84}

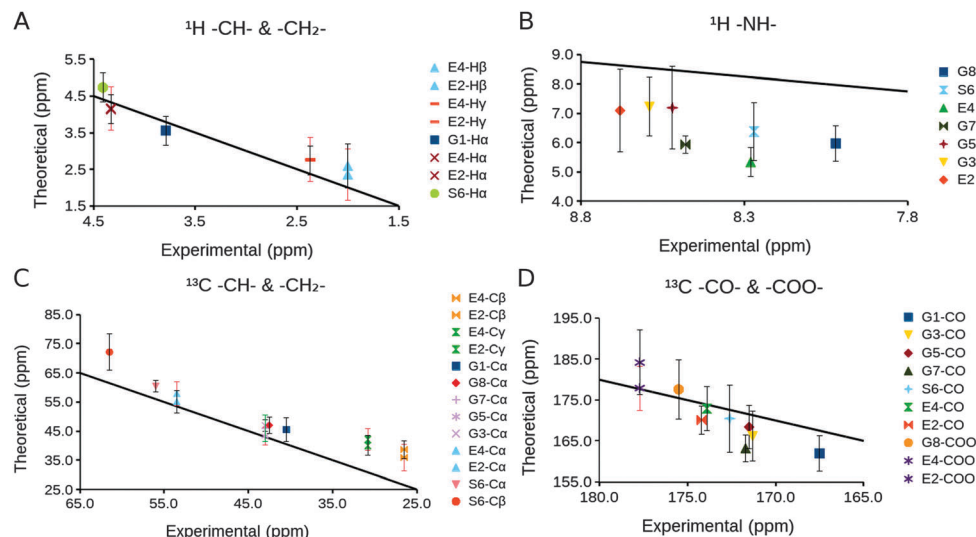


Fig. 5 The average values of the NMR chemical shifts computed with the free octapeptide. The black line is the perfect agreement between theoretical and experimental values. The bars are the standard deviations.

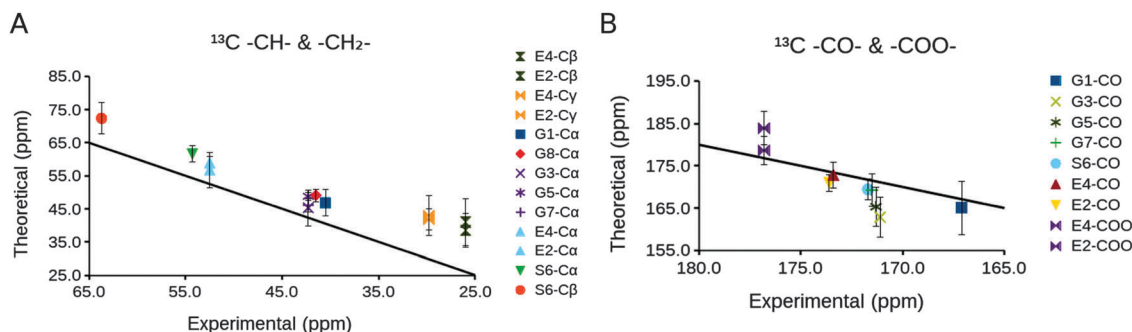


Fig. 6 The average values of the ^{13}C NMR chemical shifts computed with the phosphorylated octapeptide. The black line is the perfect agreement between the theoretical and experimental values. The bars are the standard deviations.

3.2.2. Al(III) binding to the GEGEGSGG peptide. As was mentioned above, line-broadening and loss of some of the ^{13}C NMR signals in the NMR spectra upon complexation were observed experimentally;⁴⁸ the former cannot be reproduced, but the latter can be simulated to compare the computed chemical shifts of the aluminum complexes with the free peptide, specially those of the groups involved in binding. The N1, N2 and N3 models have monodentate coordination to the carboxyl group and in N4, N5 and N6 are bonded *via* bidentate modes (Fig. 1).

The results show that the aluminum coordinated to E2 change their CO and COO values by 3.0 ppm (Fig. 7 N1), the other groups undergo changes to the same extent. The coordination to E4 displays a similar trend with variations of around 4.0 ppm for the CO and COO groups, the CO value of G7 deviates by 7.3 ppm, but it is within the range of differences obtained for the free peptide (Fig. 7 N2). Coordination to the C-terminal in a monodentate fashion displays similar variations as observed in the previous cases (Fig. 7 N3), but in combination with the carbonyl, larger deviations are found with respect to the free octapeptide values,

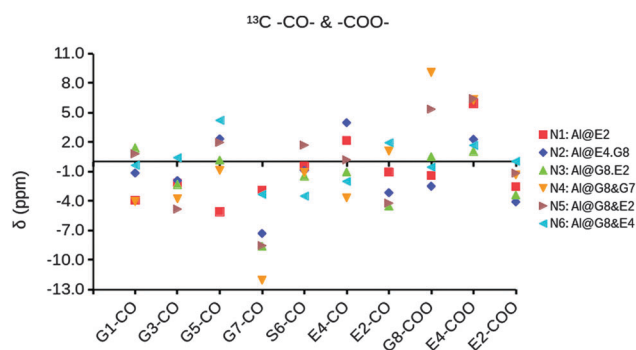


Fig. 7 The differences in the computed ^{13}C NMR chemical shifts found between the free octapeptide and its Al(III) complexes.

12.1 ppm in CO-G7 and 9.1 ppm in the COO-terminal, the other groups have similar variations as observed in the other models (Fig. 7 N4). The coordination of the C-terminal combined with any glutamic acids (Fig. 7 N5 and N6) show that the carbons involved in the coordination remain intact and are not reflected

the experimental changes, and therefore, their existence is hardly conceivable, as stated.⁴⁸

Taking into account the agreement with the experimental values in the free octapeptide and the variations in the aluminum complexes, the coordination with the glutamic acids does not reflect the changes in the experimental NMR spectra and neither does the monodentate coordination with the C-terminal; only when the carbonyl of G7 is involved in the coordination, the differences in the computed values of the carboxyl groups on G8 and E4 are greater than 6.0 ppm and twice in the carbonyl on G7, such shifting on G7 could be hidden by the signal broadening.

3.2.3. Al(III) binding to the phosphorylated peptide. In the phosphorylated spectra, the line-broadening of the ¹³C NMR signals is substantially changed, as well as the shifting of the CH and CH₂ signals from S6, which help us to distinguish between the coordination modes. The models have the aluminum bound to the phosphate group *via* a mono and bidentate manner (P1 and P2), to the C-terminal as a bidentate ligand (P3) and to the phosphate combined with the C-terminal or with E4 (P4 and P5), the monodentate coordination was not tested because similar results are expected to those obtained with the non-phosphorylated octapeptide.

The results for the differences between the aluminum complexes and the phosphorylated peptide show that the values of S6 have differences of about 3.0 ppm (Fig. 8 P1), indicating small structural changes and therefore small signal shifts. The bidentate phosphate shows similar variations to the monodentate complex and it is not possible to discard this coordination mode (Fig. 8 P2), but for the bidentate C-terminal it is, because it modifies the CO region considerably with differences greater than 7.0 ppm and the CH and CH₂ signals of S6 are slightly affected (Fig. 8 P3). The combined phosphate with the C-terminal changes its carboxyl signal by about 3.0 ppm, but the most interesting results are the changes of 8.8 and 1.2 ppm in the CH and CH₂ groups of S6, respectively (Fig. 8 P4), although the variation of CH₂ is small, this model displays the biggest change in the aliphatic region; while the combination with the carboxyl of E4 does not reflect the significant changes in the aliphatic region nor in the signals of the carbons involved in the coordination (Fig. 8 P5), the greatest variations correspond to the G3 and G5 carbonyls both with 8.0 ppm and

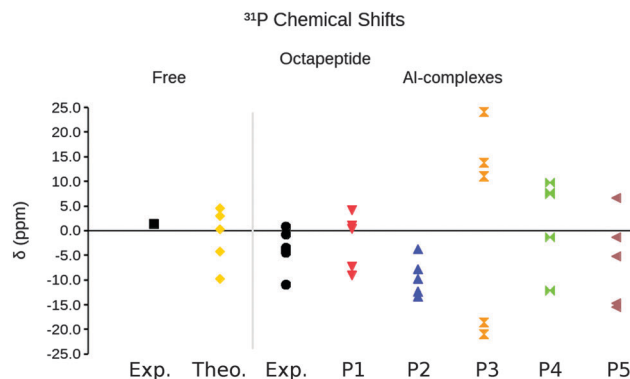


Fig. 9 The experimental and theoretical ³¹P NMR chemical shifts for the phosphorylated peptide and its aluminum complexes. The experimental values are taken from the literature.⁴⁸

experimentally, it is only observed by line-broadening in that region. Therefore, according to the ¹³C NMR chemical shifts, the best candidates used to reproduce the experimental situation are P1 and P4, wherein the aluminum coordinated to the phosphate group *via* a monodentate fashion does not alter the resonance pattern and the binding of C-terminal produces the observed signal shifts for the aliphatic carbons of serine.

Finally, aluminum coordination to the phosphate group was proved experimentally by comparing the ³¹P NMR chemical shifts of the phosphorylated peptide and its complexes; it was assigned the signal at 1.36 ppm in the free octapeptide and in the range from 0.8 to −11 ppm in the aluminum complexes.⁴⁸

The computed values of the phosphorylated peptide show positive and negative values (Fig. 9 free), averaging them we have values of 2.6 and −7.0 ppm, and a variation with the most intense experimental signal of less than 2 ppm. For the aluminum complexes, the computed values of the five models show that if the phosphate group does not participate in aluminum coordination the ³¹P NMR chemical shifts have the largest deviation from the experimental region (Fig. 9 P3), whereas the other cases are closer to the experimental trend and to discard one of them is not possible; however, the model P1 would be the best candidate to display the experimental resonance pattern.

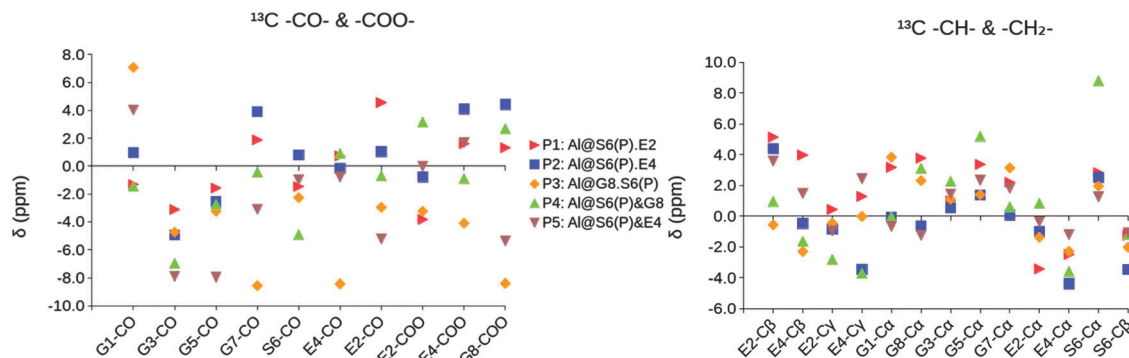
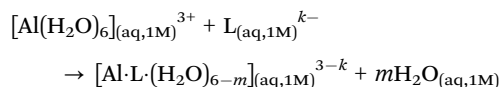


Fig. 8 The differences of the computed ¹³C NMR chemical shifts between the phosphorylated octapeptide and its Al(III) complexes.

3.3. Binding free energies in solution

The complex formation stability was studied using the next ligand substitution reaction as follows:



where L is the octapeptide or the phosphorylated derivative with $k = 2$ and 4, respectively, and m depends on the coordination mode, 1 for monodentate and 2 for bidentate. The free energy in solution corresponding to the aluminum binding to the peptides is calculated as follows:

$$\Delta G_{\text{aq}} = G_{\text{aq}}([\text{Al}\cdot\text{L}\cdot(\text{H}_2\text{O})_{6-m}]^{3-k}) + m\cdot G_{\text{aq}}(\text{H}_2\text{O}) - G_{\text{aq}}([\text{Al}(\text{H}_2\text{O})_6]^{3+}) - G_{\text{aq}}(\text{L}^{k-}) + \Delta nRT\ln(24.26) + mRT\ln(55.34)$$

where the last two terms account for the volume change due to the transformation from 1 atm to 1 M in solution, Δn indicates the change in the number of species in the reaction and the concentration of 55.34 M of water in liquid water.⁸⁵ The enthalpy values were also computed excluding the last term.

The complexation energies of both the non-phosphorylated and phosphorylated aluminum complexes are displayed in Fig. 10. The aluminum binding energy results for the non-phosphorylated complexes (**N1** to **N6** in Fig. 1) show that monodentate coordination with E2 (**N1**) would be unlikely ($-33.70/-36.48$ kcal mol⁻¹ for $\Delta G_{\text{aq}}/\Delta H_{\text{aq}}$) and that there is a clear preference for coordination to E4 (**N2**, $-60.98/-71.81$ kcal mol⁻¹) and the C-terminal carboxylate (**N3**, $-56.47/-67.19$ kcal mol⁻¹), with both sites being competitive for Al binding.

Furthermore, we also estimated the binding affinities found in the bidentate binding modes with two residues (**N5** and **N6**). In both structures, one of the ligands is through the C-terminal of G8, whereas the other coordination is through the acetate group of E2 (**N5**) or E4 (**N6**). Again, we detected a preference for E4 coordination with a binding energy of $-67.86/-69.71$ kcal mol⁻¹ for **N6** versus $-58.17/-59.57$ kcal mol⁻¹ for **N5**.

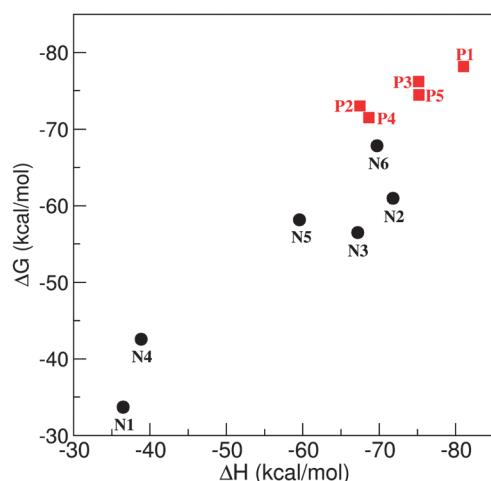


Fig. 10 The complexation energies of the non-phosphorylated (black) and phosphorylated (red) aluminum complexes.

In summary, our results for the non-phosphorylated octapeptide points out to a competition between E4 and the C-terminal for binding to aluminum, with a variety of situations of similar energy: monodentate to one of the groups or bidentate to both of them. This also suggests that it is highly likely that the octapeptide coordinates two aluminum atoms, one at the C-terminal and the other at E4 as was suggested by Hollender *et al.*,⁴⁸ but the dinuclear species are beyond the scope of this study.

On the other hand, the binding energies for the phosphorylated octapeptide were studied with five models (**P1** to **P5** in Fig. 1). In this case, it is observed that the preferred coordination mode is through the phosphate at S6 (**P1**, $-78.18/-81.04$ kcal mol⁻¹), with other groups binding to water in the first coordination shell of aluminum. There are also other competitive structures close in energy such as **P3** ($-76.21/-75.16$ kcal mol⁻¹) in which aluminum is interacting directly with the C-terminal and indirectly with the phosphate at S6 through one of the coordinating waters, and **P5** ($-74.47/-75.19$ kcal mol⁻¹) in which aluminum is interacting bidentately with S6 and E4. It is worth noting that during the optimization some proton transfer was observed from the water molecules to the closer carboxylic group (Note S1, ESI[†]), which are known to be due to limitations in the solvation models.⁸⁶

Our calculations demonstrate a clear increase in the aluminum binding affinity upon phosphorylation. Averaging over the three most favorable structures in each case (non-phosphorylated and phosphorylated octapeptide); we obtain an average binding free energy/enthalpy of $-61.8/-69.6$ kcal mol⁻¹ for the non-phosphorylated octapeptide versus $-76.3/-75.2$ kcal mol⁻¹ for the phosphorylated one. Thus, we can conclude that phosphorylation leads to a higher affinity for aluminum and therefore the presence of aluminum can favor thermodynamically the formation of phosphorylated peptides. It can be noted, however, that our calculations correspond to an acidic pH (as were the NMR experiments of Hollender *et al.*). Extension of the present results to more physiological pH values should be done with caution, because the affinity of the octapeptide towards aluminum could be altered by the presence of hydroxides in the first solvation layer around aluminum as we increase the pH.

4. Biological implications

The interest in studying the structure and thermodynamic properties of the formation of aluminum complexes with the neurofilament analogue octapeptide GEGEGSGG and its Ser(P) derivative is high due to two main reasons. On one hand, the analysis of the structural effects that aluminum provokes upon binding to this polypeptide could shed light on the mechanism by which aluminum enhances the aggregation of β -amyloids and NFTs. On the other hand, phosphorylation could be key to finding a lead compound for the design of a selective peptide-based Al(III) chelator and our data could help in the rationalization of the effect expected binding affinity upon phosphorylation.

Based on the theoretical results of the present study, we can conclude that phosphorylation has a profound effect on the binding pattern of a polypeptide to Al(III). Phosphorylation changes the structure of the binding pattern of Al(III) towards the octapeptide, shifting the preferred binding site from E4 to S6(P), with the C-terminal carboxylate being a favorable binding site in both non-phosphorylated and phosphorylated peptides. In addition, phosphorylation significantly increases the affinity of the GEGEGSGG octapeptide towards Al(III) binding. An increase of around 15 kcal mol⁻¹ in binding free energy was estimated by our calculations upon phosphorylation. Our results are coherent with the experimental NMR information and points out to the preference of coordination of Al(III) to negatively charged groups, specially phosphate monoester functional groups bearing two negative charges. Our previous estimates for the aluminum binding affinities of phosphate monoesters gave an estimate of -65.9/-67.3 kcal mol⁻¹.³¹ Our results for the phosphorylated octapeptide **P1** are significantly higher -78.2/-81.0 kcal mol⁻¹ and indicative of the collaborative effects between the phosphate and carboxylate groups in binding to aluminum.

Is the increase in binding affinity upon phosphorylation enough to transform the present octapeptide to a strong Al(III) chelator in biological systems? To answer this question, we compared our binding affinities to the ones calculated in previous studies for known strong biological chelators: citrate, the main low molecular mass chelator of aluminum in blood serum; 2,3-DPG, a chelator of aluminum in red cells, and ATP-like triphosphates.^{31,86} The binding free energies/enthalpies for these chelators are: citrate (-133.0/-124.9 kcal mol⁻¹) > 2,3-DPG (-123.5/-118.9 kcal mol⁻¹) > ATP-like triphosphates (-109.2/-108.7 kcal mol⁻¹). Thus, these chelators show much larger binding affinities than the phosphorylated octapeptide. In this sense, it is clear that the phosphorylation of one residue is not enough to transform the octapeptide into a strong Al(III) chelator. In fact, Hollender *et al.* established that neither the non-phosphorylated nor the phosphorylated peptide prevented the precipitation of Al(III) complexes at around the physiological pH, keeping Al(III) in solution only in the weakly acidic pH range.⁴⁸

On the other hand, based on the favorable formation of intramolecular bridges between functional groups through bidentate structures, we can expect that aluminum could induce the aggregation of polypeptides through different types of intermolecular bridges. The most likely possibility at low pH would be through phosphate-Al-phosphate intermolecular bridges, with the phosphates located in different polypeptides. However, other competitive structures would be carboxylic-Al-phosphate or carboxylic-Al-carboxylic bridges, which could be more likely at higher pH. In fact, in our previous study with the 2,3-DPG phosphate chelator,⁸⁶ we have demonstrated the propensity of aluminum to form 1:2 complexes and therefore the possibility to act as an aggregation agent for phosphates. Our results in the present study point in the same direction. Moreover, we have observed that the formation of such bidentate structures provokes changes in the secondary structure

of the polypeptide, increasing its stiffness and favoring the conformational space region associated to β -sheet structures.

5. Conclusions

In this study, we have applied a theoretical protocol (*vide supra*) to study the coordination of aluminum to a synthetic neurofilament analogue GEGEGSGG octapeptide, both phosphorylated and non-phosphorylated, by combining classical mechanics simulations, the selection of structures based on clustering methods, DFT evaluation of the NMR shielding constants and DFT/PCM calculations of the binding affinities. Our results shed light on the important subject of the interaction of aluminum to polypeptides, a key interaction at the center of the potential toxic effects of aluminum.

Our results point to a clear enhancement in the affinity towards aluminum upon phosphorylation of the peptides in this model system, with changes that covers both structural and thermodynamic aspects. On the one hand, phosphorylation changes the preferential binding site for aluminum, shifting the preference from C-terminal and/or G7/E4 in the non-phosphorylated octapeptide to S6(P) upon phosphorylation, either monodentately or in combination with the C-terminal or E4. Our calculations also show a variety of possible structures for aluminum binding, suggesting that there are a variety of competitive structures for aluminum binding, which leads to a rich diversity of potential structures to be formed. Our data is in agreement with the previous NMR estimates and therefore allows for a clarification of the previous experiments. Whether this behavior can be extended to peptides of larger size and to proteins in general, will be the subject of future study.

On the other hand, the binding of the polypeptide to aluminum could have a sizable effect on its secondary structure, which is sensitive to the type of coordination to the metal. Thus, especially in the case of bidentate structures, in which aluminum is bound to two different groups of the octapeptide, we have observed an enhancement of the population of the regions associated with β -sheet structures in the Ramachandran plots, with an increase in the stiffness of the backbone.

Finally, the computation of the affinity binding energies clearly shows a promotion of aluminum binding upon peptide phosphorylation. This enhancement is not enough to transform the GEGEGSGG peptide into a strong aluminum chelator. Our results clearly point to the possibility of the promotion of protein aggregation induced by aluminum *via* the formation of intermolecular phosphate monoester bridges.

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